SARAH MUHAMMAD RASHAD

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EDUCATION

Ain Shams University, Faculty of Engineering Cairo, Egypt

Bachelor of Science in Electrical Engineering (Electrical Power and Machines Department)

- Expected Graduation: July 2027
- Grade: Good

TECHNICAL SKILLS

- **Programming:** C (Advanced), C++, Python (Basics), Arduino, Git/GitHub
- **Simulation & Control:** MATLAB, Simulink, Simscape Battery, Proteus, SimulIDE
- **Power Systems & Design:** ETAP, PowerWorld Simulator, PSAT, DIALux evo, Ecodial INT, AutoCAD, PCB Design (EasyEDA)
- **Soft Skills & Leadership:** Team Leadership, Technical Instruction, Workshop Facilitation, Adaptability
- **Languages:** Arabic (Native), English (Good working proficiency)

TECHNICAL PROJECTS

DC-DC Cuk-Converter Design & Simulation Dec 2025

- Modeled and simulated a DC-DC Cuk converter circuit to optimize power conversion efficiency and ripple suppression.

Solar Tracking Systems (SCC & Arduino-Based) Apr 2024 – Dec 2024

- Developed a pure hardware implementation using **Signal Conditioning Circuits (SCC)** and analog components for automated solar tracking without microcontrollers.
- Built a microcontroller-based dual-axis tracking system using **Arduino Uno** and light-dependent resistors (LDRs).

ATmega32 Embedded Drivers & Peripheral Interface Dec 2023 – Feb 2024

- Implemented bare-metal peripheral drivers (GPIO, EXTI, LCD, Keypad) from scratch using C for ATmega32 microcontrollers.

Library Management System (C / Data Structures) Aug 2023

- Programmed a CLI-based management tool; transitioned architecture from initial arrays to a doubly linked list algorithm for optimized memory handling.

Smart Garage System (Arduino Uno) Feb 2023

- Awarded 1st Place at BETA Academy Robotics Workshop (Round 14) for developing an automated entry/exit parking system using IR sensors and servo motors.

TRAININGS

General Authority for Educational Buildings Training Aug 2026

ABB — Bright Minds Training Program Jul 2025 – Aug 2025

Cairo Electricity Production Company Training Sep 2024

The Arab Contractors Training Aug 2024

LEADERSHIP & STUDENT ACTIVITIES

Athar (أثر) Student Activity for Girls Jun 2026 – Present

Founder & President

- Founded a student initiative focused on empowering female engineering students.
- Led event planning, operational strategy, and team recruitment from the ground up.

SemiColon Student Activity Jul 2024 – Aug 2024

C / Embedded C Workshop Leader & Instructor

- Led the C/Embedded C workshop, personally delivering 4 of 12 sessions and supervising the whole workshop.
- Evaluated all participants' final projects and gave each one detailed individual feedback.

CERTIFICATIONS & COURSES

- **MATLAB, Simulink & Simscape Battery Onramps** MathWorks (2025 – 2026)
- **Model-Based Design Concepts, Code Generation & AC/DC Converter in Simulink** OS Academy (2025 – 2026)
- **PCB Design Essentials** Mahara-Tech (Apr 2025)
- **Python Basics: Selection and Iteration** Codio (2025)
- **Introduction to AI** Almadrasa (2025)
- **EU 2025 Phase 1 Technical Program** Schneider Electric (2025)